Quantitative Inversion of Oil Film Thickness Based on Airborne Hyperspectral Data Using the 1DCNN_GRU Model

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***Abstract*— Oil film thickness (OFT) is an important indicator for estimating the amount of oil spill, and accurately quantifying the OFT is of great significance for loss assessment. In this article, hyperspectral images (HSIs) of different OFTs (0.01–3.04 mm) through a ground experiment were obtained, and the spectral characteristics were analyzed. To address the issue of poor spectral separability for different OFTs, the 1-D convolutional neural network_gate recurrent unit (1DCNN_GRU) model was developed for the quantitative inversion of OFT. It was validated through experiments on airborne Cubert-S185 HSI. The experimental results indicated that: 1) the proposed 1DCNN_GRU model effectively addressed the issue of reduced quantitative inversion accuracy resulting from poor spectral separability. The inversion results of it outperformed those of the support vector regression (SVR), convolutional neural network (CNN), and gate recurrent unit (GRU) models. Moreover, the optimal time for hyperspectral sensor to monitor OFT was at noon. 2) The proposed model using airborne hyperspectral data exhibited excellent inversion performance for OFT greater than 0.07 mm, especially with the best performance in 0.60–0.90 mm. 3) The accuracy of HSI-based OFT inversion assisted by brightness temperature (BT) data was superior to that of OFT inversion using single-source data. In particular, the proposed model had advantages in the feature level and decision level inversion of OFT in the ranges of 0.01–0.30 and 1.00–3.04 mm, respectively. This research provides technical support for the detection of OFT.**

***Index Terms*— 1-D convolutional neural network_gate recurrent unit (1DCNN_GRU), hyperspectral image (HSI), oil film thickness (OFT), quantitative inversion.**

Manuscript received 10 August 2023; revised 23 September 2023 and 10 October 2023; accepted 16 October 2023. Date of publication 19 October 2023; date of current version 31 October 2023. This work was supported in part by the National Natural Science Foundation of China under Grant U1906217, Grant 61890964, Grant 42206177, and Grant 42076182; in part by the Shandong Provincial Natural Science Foundation under Grant ZR2022QD075; in part by the Fund of Technology Innovation Center for Ocean Telemetry, Ministry of Natural Resources, under Grant 2022004; in part by the Qingdao Postdoctoral Application Research Project under Grant qdyy20210082; and in part by the Fundamental Research Funds for the Central Universities under Grant 21CX06057A. *(Meiqi Wang and Junfang Yang contributed equally to this work.) (Corresponding author: Junfang Yang.)*

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Digital Object Identifier 10.1109/TGRS.2023.3325805

I. Introduction

In RECENT years, due to the continuous expansion of offshore oil exploration and transportation, oil spill incidents at sea have become more frequent, causing significant harm to the marine ecological environment and human society [1], [2], [3], [4]. Estimating the amount of oil spill requires determining the spatial distribution and oil film thickness (OFT). Accurately quantifying the OFT on the sea surface can provide a decision-making basis for relevant departments in emergency response situations, and is of great significance for assessing the losses caused by oil spills [5], [6], [7], [8]. Remote sensing technology can monitor the dynamic changes of targets and has a large monitoring range, making it widely used in monitoring oil spills on the sea surface [5], [9], [10], [11].

Different remote sensing technologies have their unique advantages and applicability in monitoring oil spills on the sea surface [12]. For example, radar remote sensing technology has the ability to penetrate clouds and fog, enabling all-weather and all-time oil spill monitoring [13]. However, due to the similarity in radar wave signals reflected by different OFTs, it is challenging to distinguish them [14]. Optical remote sensing has shown excellent performance in oil spill monitoring. Among them, thermal infrared remote sensing technology can quantitatively invert the OFT by measuring the temperature difference between the oil film and the surrounding seawater [15], [16], [17], [18], [19]. However, meteorological conditions can significantly impact temperature differences, affecting the measurement results of OFT [20]. Ultraviolet (UV) remote sensing technology is highly sensitive to extremely thin oil films and can detect OFT smaller than 0.05 μm. However, the detection range of UV is limited and is generally only effective at short distances [21], [22]. Laser remote sensing technology can quantitatively invert the OFT

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by measuring the fluorescence spectrum of the oil film and the surrounding seawater. However, it requires high costs and significant investment in manpower and material resources [23], [24], [25].

Hyperspectral remote sensing technology has the advantages of high spatial resolution and high spectral resolution, allowing for fast and accurate detection and quantitative analysis of oil films on the sea surface, making it an important means of oil spill monitoring [26], [27], [28], [29]. By analyzing the spectral characteristics of the oil film, it is possible to determine the spatial distribution of the oil spill, the type of oil spilled, and the OFT [30], [31], [32], [33]. Quantifying OFT is an essential application direction of hyperspectral remote sensing technology in offshore oil spill monitoring. In recent years, it has been widely used for the quantification of OFT, focusing mainly on the spectral characteristics analysis and inversion methods of different OFTs [26], [30], [31], [32], [33], [34].

Regarding spectral analysis of OFT, many scholars have conducted extensive research. In the analysis of the spectral characteristics of oil films, studies have shown that different OFTs have different spectral reflection characteristics in the hyperspectral range [26], [30], [31], [33]. For instance, Jiang et al. [35] conducted an outdoor experiment to obtain hyperspectral images (HSIs) of three thicknesses of crude oil films, and found that the total remote sensing reflectance was inversely proportional to the OFT. The spectral reflection curve of the oil film has clear absorption peaks and fluctuations, and as the thickness of the oil film increases, the position and intensity of the absorption peaks and fluctuations also change. These spectral reflection characteristics provide an important basis for the quantitative inversion of OFT.

In terms of quantitative inversion methods for OFT, model-based inversion and data-driven inversion methods are mainly used. Model-based inversion methods typically use radiative transfer or reflectance models. Based on the optical characteristics of oil films, OFT can be quantitatively inverted by simulating the difference between the oil film reflectance spectrum and the observed spectrum [30], [31], [32], [34]. Data-driven inversion methods typically use machine learning algorithms to accurately invert OFT through a training sample library and feature extraction methods [36], [37], [38], [39]. For example, machine learning algorithms that have been widely used in recent years include support vector machine (SVM), artificial neural network (ANN), random forest (RF), and deep learning [40]. These methods do not require accurate physical models and directly learn the OFT inversion model from the data using a data-driven approach, which has higher robustness and prediction accuracy. These studies provide new ideas and methods for using hyperspectral remote sensing technology for offshore oil spill monitoring and emergency response.

In general, hyperspectral remote sensing technology has made significant strides and accomplishments in the inversion of OFT, but still faces a number of challenges and issues. For instance, the quality of hyperspectral remote sensing data has a significant impact on the accuracy and reliability of OFT inversion, and the quantitative inversion accuracy of OFT requires further improvement. Additionally, machine learning algorithms need to be further optimized and improved to enhance their effectiveness in this regard. To address these challenges, this article designed a ground experiment to obtain airborne HSIs of 18 different OFTs, with the aim of studying the spectral characteristics of different OFTs. The 1-D convolutional neural network_gate recurrent unit (1DCNN_GRU) model proposed in this article was employed for quantitative inversion of OFT, and the optimal range of OFT detectable by airborne hyperspectral sensor was discussed. Furthermore, the article explored the ability of combining hyperspectral and thermal infrared technology to invert OFT at both the feature level and decision level.

II. Data and Methods

A. Data Acquisition and Processing

1) Experiment: The objective of this experiment was to obtain airborne HSIs of various OFTs at different time intervals, and to investigate the spectral reflection characteristics of different OFTs, as well as the capabilities of hyperspectral sensor in retrieving OFT. The experiment was conducted on September 6, 2022, with clear weather conditions and wind speeds ranging from 0.33 to 4.08 m/s. The experiment was conducted in a black rectangular water tank filled with seawater. Eighteen experimental rings (diameter: 7 cm) were uniformly placed in the water tank, with each ring containing different OFT ranging from 0.00 to 3.04 mm. Crude oil (component analysis: wax: 7.56; gum: 18.77; asphaltene: 1.57 wt%; and freezing point: −9 °C) from the Shengli Oilfield in Dongying was used for the experiment, and the density of the crude oil was measured at different temperatures under laboratory conditions. Due to the low volatility of crude oil, changes in thickness during the observation period were disregarded.

The airborne Cubert-S185 hyperspectral imaging system was employed in this experiment, which has a spectral range of 450–950 nm, 125 bands, and an imaging size of 1000 × 1000 pixels. To record meteorological data, a Kestrel 5000 wind speed meter (measurement accuracy: 0.1 km/h) was utilized during the experiment. The experimental setup is depicted in Fig. 1.

During the experiment, after a period of static settling, the oil film was measured multiple times using an infrared thermometer at different positions on the circular ring. If the temperatures approached similarity, a portable thermal infrared imager was used to capture an image. If the colors within each ring appeared the same, except for the shadowed areas, it was considered as an indication of uniform diffusion of the oil film at that time. The radiometrically calibrated airborne Cubert-S185 hyperspectral imaging system was utilized to capture HSIs of the various OFTs at 11:00, 13:00, and 15:00. The drone was flown at a height of 10 m during image acquisition, resulting in a spatial resolution of approximately 4 mm. Ten repetitions were taken at each moments, resulting in a total of 30 HSIs.

2) Data Processing: The experimental data collected consisted of the radiation brightness of various OFTs. To begin

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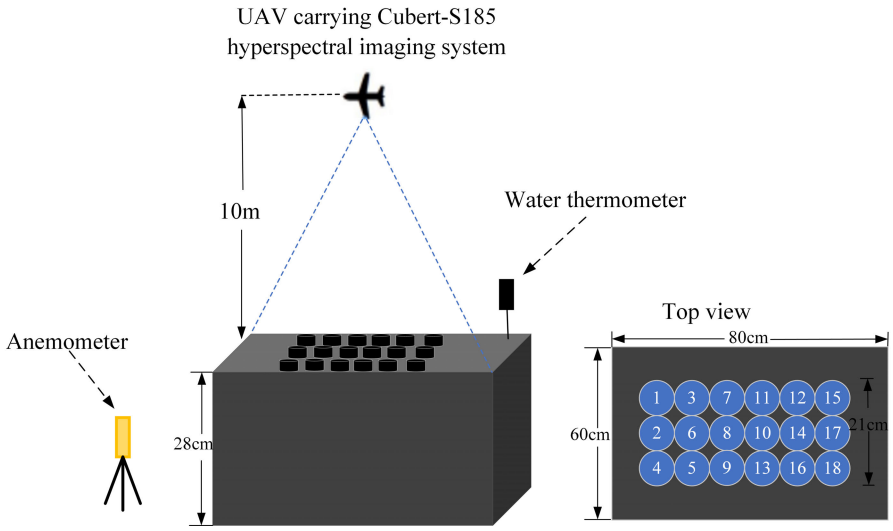


Fig. 1. Schematic of OFT monitoring scenario based on airborne hyperspectral imaging system.

with, it was necessary to perform radiation calibration and remote sensing reflectance calculation using the CubertUtilsTouch platform and MATLAB. On the CubertUtilsTouch software, followed these steps. 1) Set the mode to reflectance. 2) Chose 100% for pan sharpen. 3) Selected the multichannel TIFF format. (4) Chose the valid spectral bands in the channels option, such as "450–950." (5) After completing the settings, exported the data. The exported data represented reflectance values scaled by a factor of 10 000, which was converted into reflectance values using MATLAB. The calculation formulas used were as follows:

$$R_{\mathrm{rs}} = \frac{L_o}{\pi . L_b} \tag{1}$$

where R_{rs} represents the remote sensing reflectance of different OFTs. L_o represents the radiation brightness data of different OFTs. And L_b represents the radiation data of the standard board (AZ-WS20) used for radiation calibration.

Following preprocessing of the HSIs, spectral reflectance data for the various OFTs were obtained. ArcGIS software was utilized for image cropping and data sample labeling. The principle of cropping was to reduce the area of water in the black rectangular water tank while ensuring that all rings were included. Owing to the deviation of the shooting angle, the sizes of the images cropped at the three moments were not precisely the same. Experimental labeling was based on the measured OFT map obtained at the experimental site and the captured HSI, with each thickness containing approximately 240 pixels. The cropped HSI and data labels can be seen in Fig. 2.

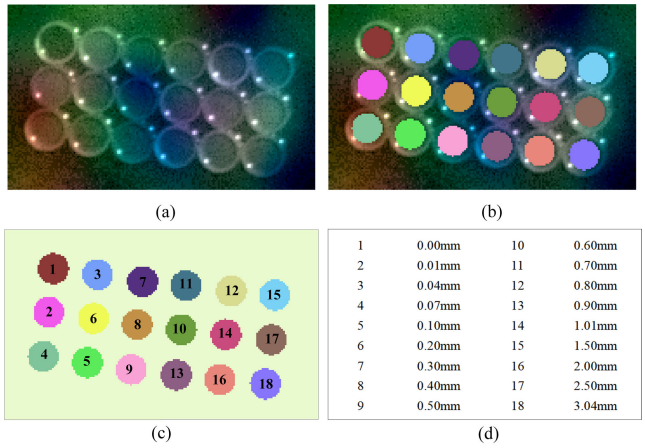


Fig. 2. (a) Hyperspectral true color image at 13:00 (image size: 177 × 107). (b) Data labels (color circular region: experimental samples). (c) Corresponding OFT number of the experimental sample. (d) OFT corresponding to the OFT number.

B. Basic Inversion Model Construction

The data utilized in this experiment consisted of airborne HSIs. Due to the small size of the experimental scene and the oil film covering the entire surface of the annulus, spatial feature information was not taken into account. Instead, a mapping relationship was established between the spectral reflectance data of various OFTs and OFT data. To this end, three basic models were selected to perform the inversion of OFT. The first was the support vector regression (SVR) model, representing the machine learning model, followed by the 1DCNN model in the convolutional neural network (CNN), and the gate recurrent unit (GRU) in the recurrent neural network (RNN).

The input data for all models was the spectral reflectance data of various OFTs, with the output data being the measured OFT data. Each thickness of the experimental sample was randomly divided in a 6:4 ratio, with 60% used for training and 40% utilized for testing. The partitioned dataset was fixed and utilized for subsequent experiments. All experimental results were averaged over five results.

1) SVR: The SVR model is a regression model used by SVM which is based on an insensitive loss function. It maps the input data to a high-dimensional space through nonlinear mapping, and establishes a linear model in this feature space to estimate the regression function [41]. During multiple training iterations, it was discovered that the kernel function used

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in the model was set to radial basis function (RBF), with the insensitive loss function set to 0.01. The optimal gamma function and penalty factor c were automatically optimized using the libSVM function in the MATLAB environment [41], resulting in the model having the most effective performance.

2) 1DCNN: The 1DCNN model is a typical CNN model with strong feature extraction capabilities. Since the data used in the experiment is pixel-level spectral reflectance data, the 1DCNN model was selected, with the kernel sliding along 1-D. Compared to traditional convolution, 1DCNN has a receptive field based on the 1-D convolution kernel, resulting in stronger feature extraction capabilities [35], [42]. The model consists of input–output, and hidden layers, with the hidden layer consisting of two convolutional layers, ReLU layers, and two fully connected layers, and the output layer set as the regression layer. After multiple rounds of training for parameter tuning, it was discovered that the first layer convolution kernel size was set to 3, with the number of convolution kernels at 4. The convolution kernel size of the second layer was set to 3, with the number of convolution kernels at 8. The number of the first and second fully connected layers was set to 20, and the number of batch training was 24. The maximum number of iterations was set to 500, and the learning rate was set at 0.01. This model was found to have the best performance.

3) GRU: RNNs are known to perform well on data with sequence characteristics. Since the input data in the experiment is the spectral reflectance data of different OFTs, which has certain sequence characteristics, the GRU model was selected. GRU is an improved model of long short-term memory (LSTM), which integrates the forget gate and the input gate into a single update gate, while mixing the neuron state and the hidden state. This model can effectively alleviate the problem of gradient vanishing in RNNs, and can reduce training parameters while maintaining the training effect [43], [44]. The network structure consists of a sequence input layer, GRU layer, dropout layer, fully connected layer, and regression layer. The purpose of the dropout layer is to prevent the model from overfitting. After parameter optimization, the best training results of the model were obtained when max epochs were set to 500, initial learn rate was set to 0.005, and mini batch size was set to 128.

C. 1DCNN_GRU for OFT Inversion

When conducting OFT inversion, the limited separability of spectral reflectance curves for oil films of varying thicknesses can result in the basic inversion model producing errors, such as misclassifying thin oil films as thick ones or vice versa. Given the real-time nature of oil spill response, it is essential to develop a model that can monitor OFT in situations where optimal detection time is not achievable. The 1DCNN_GRU model proposed in this article is obtained by integrating the 1DCNN model with the GRU model. It mainly utilizes the 1DCNN model for its stronger feature extraction capabilities in comparison to traditional convolution, while also utilizing the long-term cumulative learning ability of the GRU model. The features extracted by 1DCNN are used as input data for the GRU model to achieve deeper feature extraction. The model was validated on hyperspectral data. The network model structure can be seen in Fig. 3.

The proposed model utilized the linear rectifier function (ReLU) as its activation function. The primary objective is to enhance the model's nonlinear fitting capabilities and mitigate the issue of gradient vanishing. The formula for computing the ReLU function is as follows:

$$g(x) = \max(0, x). \tag{2}$$

The model implemented maximum pooling in its pooling layer. This layer is utilized to reduce the size of the model, increase calculation speed, and enhance the robustness of the extracted features. The formula for max pooling is as follows:

$$c_j = \max_{N\times 1}\left(c_i^{n\times 1} v(n, 1)\right) \tag{3}$$

where c_j denotes the maximum value in the neighborhood, max() represents the max pooling function, and v(n,1) is the window function for the convolutional layer.

A GRU layer is a type of RNN layer that models temporal dependencies in time series and sequential data. The layer's hidden state at time step t represents the output of the GRU layer at that time step. At each time step, the layer adds or removes information from the state using a set of gates that regulate the flow of information. The following components are responsible for controlling the hidden state of the layer: Reset gate (r), Update gate (z), Candidate state ($\tilde{h}$). The parameters that are learned in a GRU layer include the input weights W (InputWeights), the recurrent weights R (RecurrentWeights), and the bias term b (Bias) [43]

$$W = \begin{bmatrix} W_r \\ W_z \\ W_{\tilde{h}} \end{bmatrix}, R = \begin{bmatrix} R_r \\ R_z \\ R_{\tilde{h}} \end{bmatrix}. \tag{4}$$

The bias vector in a GRU layer is dependent on the ResetGateMode property. When the ResetGateMode is set to after-multiplication or before-multiplication, the bias vector is composed of three concatenated vectors. If it is set to recurrent-bias-after-multiplication, then the bias vector is composed of six concatenated vectors [43]

$$b = \begin{bmatrix} b_{Wr} \\ b_{W_z} \\ b_{W_{\tilde{h}}} \end{bmatrix}, b = \begin{bmatrix} b_{Wr} \\ b_{W_z} \\ b_{W_{\tilde{h}}} \\ b_{Rr} \\ b_{R_z} \\ b_{R_{\tilde{h}}} \end{bmatrix}. \tag{5}$$

The hidden state at time step t is calculated as follows [43]:

$$h_t = (1 - z_t) \odot \tilde{h}_t + z_t \odot h_{t-1}. \tag{6}$$

The following equations describe the computations that occur at time step t:

Reset gate	after-multiplication before-multiplication recurrent-bias-after-multiplication
Update gate	after-multiplication before-multiplication
Candidate state	recurrent-bias-after-multiplication after-multiplication

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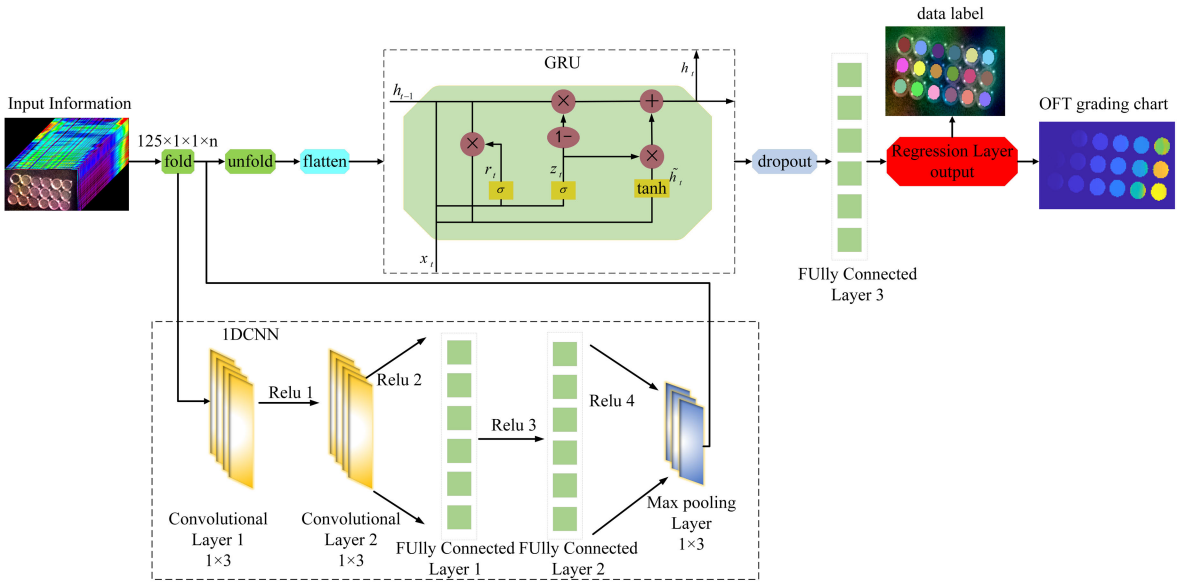


Fig. 3. Network model structure of 1DCNN_GRU.

In these computations, σ_g and σ_s represent the activation functions for the gate and state, respectively. By default, the gruLayer function uses the sigmoid function given by $\sigma_x = (1+e^{-x})^{-1}$ to calculate the gate activation function, and the hyperbolic tangent function (tanh) to calculate the state activation function.

Due to the limited number of training samples, a dropout-Layer was added after the GRU layer to prevent overfitting caused by excessive network model parameters. By connecting the final fully connected layer to the regressionLayer, we achieved quantitative inversion of the OFT.

To facilitate comparison with the basic model proposed in Section II-B, the input data for the current model was the spectral reflectance data of various OFTs, while the output data was the measured OFT data. The training and test sets remained consistent with the aforementioned model.

D. Accuracy Evaluation Indicators

To quantitatively evaluate the OFT inversion results, this study compared the proposed model with three basic models and explored the optimal OFT detection range in hyperspectral imaging. Four accuracy evaluation metrics were employed in this article, namely the root-mean-square error (RMSE), mean absolute error (MAE), mean relative error (MRE), and coefficient of determination (R^2).

$$r_t = \sigma_g\left(W_r x_t + b_{W_r} + R_r h_{t-1}\right)$$
$$r_t = \sigma_g\left(W_r x_t + b_{W_r} + R_r h_{t-1} + b_{R_r}\right) \quad (7)$$
$$z_t = \sigma_g\left(W_z x_t + b_{W_z} + R_z h_{t-1}\right)$$
$$z_t = \sigma_g\left(W_z x_t + b_{W_z} + R_z h_{t-1} + b_{R_z}\right) \quad (8)$$
$$\tilde{h}_t = \sigma_s\left(W_{\tilde{h}_1} x_t + b_{w_{\hat{h}}} + r_t \odot (R_h h_{t-1})\right)$$
$$\tilde{h}_t = \sigma_s\left(W_{\tilde{h}_1} x_t + b_{W_k} + R_{\tilde{h}}(r_t \odot h_{t-1})\right)$$
$$\tilde{h}_t = \sigma_s\left(W_{\kappa_i} x_t + b_{W_i} + r_t \odot \left(R_h h_{t-1} + b_{R_{\tilde{i}}}\right)\right) \quad (9)$$

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2}{\mathrm{n}}} \quad (10)$$
$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \hat{y}_i\right| \quad (11)$$
$$\mathrm{MRE} = \frac{1}{n}\sum_{i=1}^{n}\frac{\left|y_i - \hat{y}_i\right|}{\hat{y}_i} \quad (12)$$
$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2}{\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2} \quad (13)$$

where Y is the inversion OFT value (mm). $\hat{y}_i$ is the measured OFT value (mm). $\bar{y}$ is the average value of the inversion OFT (mm). n is the number of test samples. RMSE, MAE, and MRE are used to indicate the degree of difference between the inversion and measured values, with smaller values indicating higher inversion accuracy. On the other hand, R^2 reflects the fitting performance of the model, with larger values indicating better model fitting.

III. Results and Analysis

Section III-A analyzed the spectral reflectance curves of airborne HSIs obtained from experimentation for different OFTs. The analysis focused on the absorption and reflection characteristics, specifically the variations in spectral reflectance between different OFT, providing a theoretical basis for subsequent OFT inversions. Section III-B involved a comparative analysis of OFT inversion results using the fundamental OFT inversion model. In Section III-C, the model developed in this article was utilized to conduct OFT inversions, with the inversion results being compared and analyzed alongside those obtained via the fundamental inversion model.

A. Spectral Characteristics Analysis of Different OFTs

To present the experimental findings more clearly, the 18 OFTs were categorized into three groups based on thickness,

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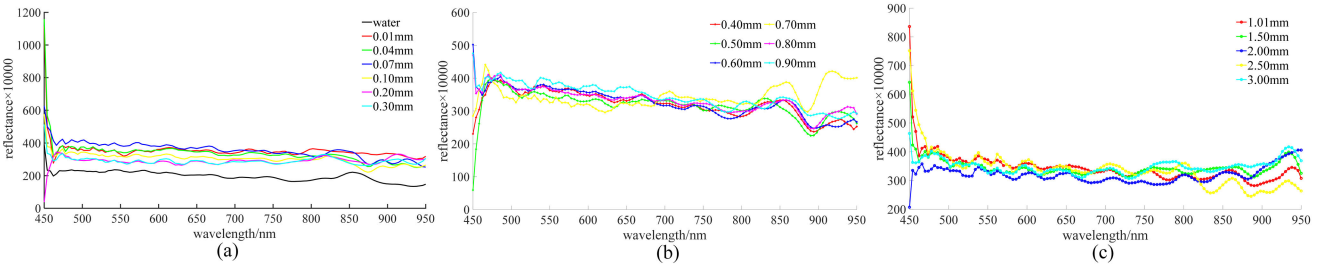


Fig. 4. Spectral reflectance curves at 11:00 for different OFTs. (a) 0.01-0.30 mm, (b) 0.40-0.90 mm, and (c) 1.01-3.04 mm.

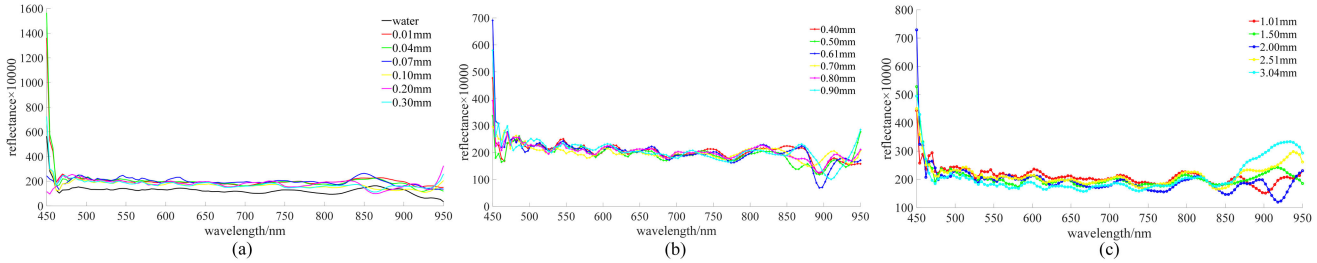


Fig. 5. Spectral reflectance curves at 13:00 for different OFTs. (a) 0.01-0.30 mm, (b) 0.40-0.90 mm, and (c) 1.01-3.04 mm.

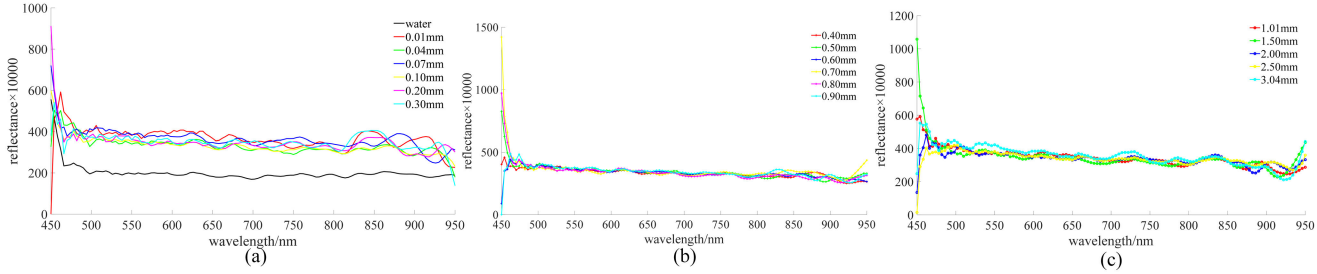


Fig. 6. Spectral reflectance curves at 15:00 for different OFTs. (a) 0.01-0.30 mm, (b) 0.40-0.90 mm, and (c) 1.01-3.04 mm.

the first group consisted of OFT ranging from 0.00 to 0.30 mm, the second group from 0.40 to 0.90 mm, and the third group from 1.00 to 3.04 mm. The spectral reflectance curves of the different OFTs were plotted accordingly (Fig. 4. for 11:00, Fig. 5. for 13:00, and Fig. 6. for 15:00). It is worth noting that the spectral reflectance curves were smoothed in this process.

For OFT ranging from 0.01 to 0.30 mm, the spectral reflectance curves in Figs. 4(a)–6(a) showed that the spectral reflectance in the visible and near-infrared bands decreases with increasing OFT. This phenomenon was primarily attributed to the oil film covering the water surface: crude oil has a strong light-absorbing ability, and this effect becomes more pronounced as the OFT increases [45], [46]. However, the similar OFT set-up used in our experiment did not exhibit an inverse relationship, likely due to the limited sensitivity of the airborne hyperspectral imaging system we employed, which cannot differentiate between similar OFTs. Upon comparing the spectral reflectance curves at the three different moments, we observed that the spectral separability of different OFTs weakened as the intensity of sunlight decreased [35]. Furthermore, we found that the reflectivity of water was less than that of water covered by an oil film, which contradicted the conclusions of previous studies [35], [45]. After examining the spectral reflectance data at various moments, we ruled out the influence of the sun azimuth and zenith angle, and concluded that this discrepancy was primarily due to the experimental conditions we set up and the effect of the tank bottom on water reflection.

The OFT in our experiment was less than 0.70 mm. In the visible light band, we observed that the spectral reflectivity decreased as the thickness of the oil film increased, as shown in Figs. 4(b)–6(b). However, in the near-infrared band, the opposite conclusion was reached. When considering Figs. 4(c)–6(c), it was observed that for OFTs ranging from 0.70 to 3.00 mm, the spectral response law did not appear to be significant. This is likely due to the limitations of hyperspectral imaging in detecting OFTs, as it is not sensitive to oil films with a thickness greater than 0.70 mm.

We elevated the spectral reflectance curve of 11:00 by 50, as depicted in Fig. 7(a)-(c). For different OFTs, comparable reflection peaks and absorption valleys were observed across most wavelengths of visible light and near-infrared. Consequently, spectral separability and a consistent trend in the spectral reflectance curves were identified among the various OFTs. These findings suggest that OFT detection through hyperspectral data is viable. However, none of the individual or combined spectral bands exhibited discriminative characteristics for distinguishing varying thicknesses of oil

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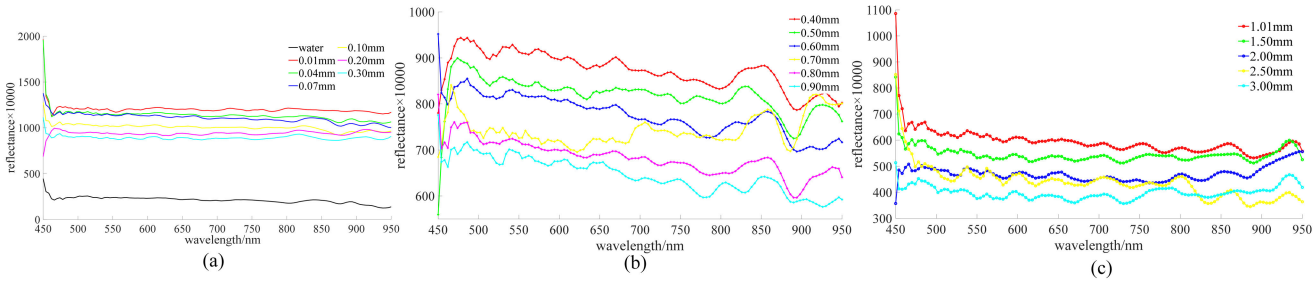


Fig. 7. Spectral reflectance curves for different OFTs at 11:00 after shifting. (a) 0.01-0.30 mm, (b) 0.40-0.90 mm, and (c) 1.01-3.04 mm.

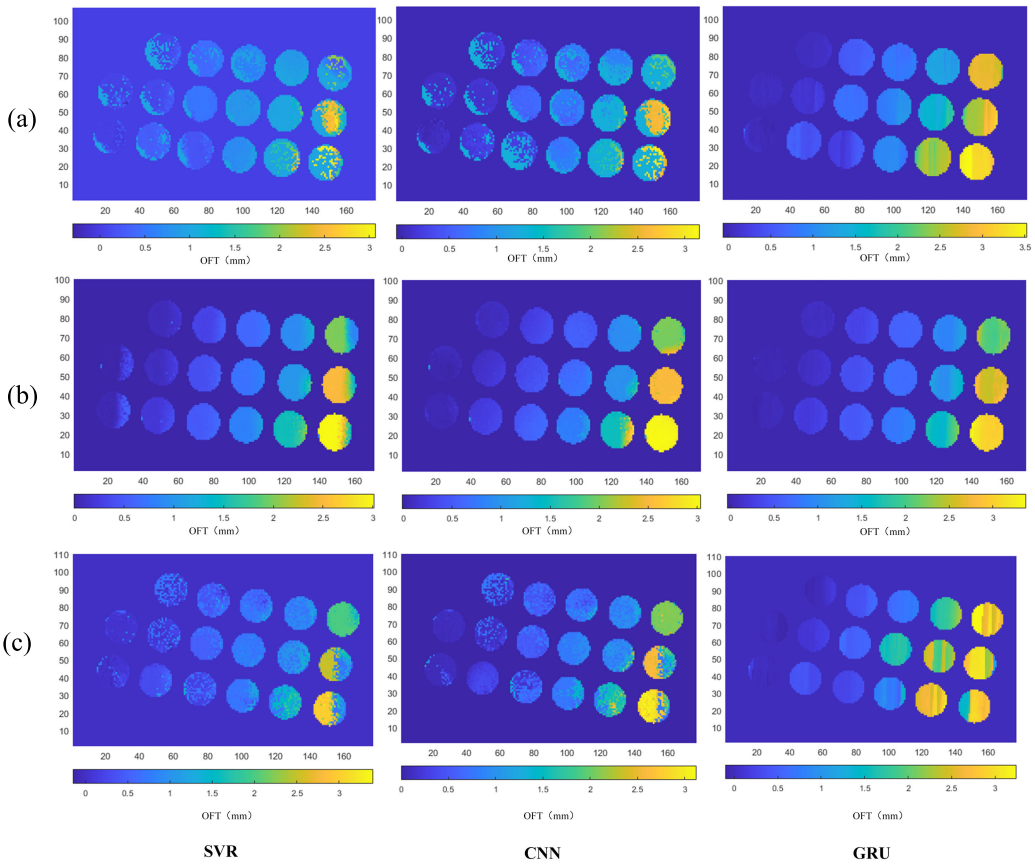


Fig. 8. OFT classification charts corresponding to the inversion results from the three models at (a) 11:00, (b) 13:00, and (c) 15:00.

films. To this end, we preserved the spectral reflectance data in its entirety for subsequent OFT inversion analysis.

B. Inversion Results Analysis of Basic OFT Inversion Model

1) Qualitative Analysis of the Inversion Results: As the experiment employed spectral reflectance data of the oil film, we opted for three fundamental inversion models to estimate the OFT. Qualitative analysis of the inversion results was performed by generating OFT classification charts based on the inversion outputs. Combined with the analysis of the OFT classification chart, it was observed that in the OFT classification chart at 11:00 [see Fig. 8(a)], the SVR model's inversion results were severely misclassified. Below an OFT thickness of 0.30 mm, there was a tendency to misclassify the thin oil film as a thicker oil film, with some OFTs even inverted to negative values. Additionally, for oil films thicker than 0.30 mm, the thickness separability was poor. The CNN model's inversion results also misclassified OFTs before 0.30 mm thickness, and at the ring's edge, the thinner oil film was sometimes inverted as a thicker oil film, leading to poor OFT separability. Conversely, the GRU model performed well for OFT less than 0.30 mm. It was evident that different OFTs exhibited distinct color changes, but for thicker oil films, the inversion values were larger, with a maximum inversion thickness exceeding 3.50 mm.

Compared to the OFT grading chart at 11:00, the inversion effect of 13:00 was superior. The SVR model's OFT grading chart [see Fig. 8(b)] at 13:00 was not clearly differentiating the thinner oil films. Additionally, misclassification occurred for OFT of 0.01 and 0.04 mm, with some OFTs being misclassified as thicker oil films, while for thicknesses greater than 2.00 mm, some thicker oil films were misclassified as

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TABLE I
ACCURACY OF THE INVERSION RESULTS USING THE BASIC MODEL

OFT / Model	Accuracy Evaluation Indicators	11:00			13:00			15:00		
		SVR	CNN	GRU	SVR	CNN	GRU	SVR	CNN	GRU
0.01-0.30mm	RMSE/mm	0.024	0.022	0.029	0.016	0.015	0.012	0.037	0.034	0.025
	MAE/mm	0.010	0.012	0.019	0.005	0.003	0.008	0.018	0.019	0.019
	MRE	0.195	0.195	0.272	0.113	0.085	0.222	0.462	0.528	0.318
	R^2	0.942	0.948	0.879	0.974	0.978	0.985	0.845	0.872	0.939
0.40-0.90mm	RMSE/mm	0.063	0.054	0.098	0.031	0.011	0.022	0.103	0.088	0.074
	MAE/mm	0.027	0.030	0.078	0.020	0.005	0.014	0.060	0.047	0.055
	MRE	0.041	0.047	0.109	0.033	0.009	0.023	0.097	0.079	0.078
	R^2	0.848	0.891	0.387	0.965	0.996	0.983	0.416	0.616	0.714
1.00-3.04mm	RMSE/mm	0.448	0.323	0.275	0.095	0.037	0.082	0.440	0.310	0.432
	MAE/mm	0.171	0.152	0.231	0.046	0.015	0.058	0.194	0.134	0.322
	MRE	0.070	0.083	0.120	0.026	0.018	0.029	0.101	0.086	0.147
	R^2	0.520	0.742	0.754	0.980	0.997	0.985	0.530	0.757	0.554

thinner oil films. The inversion results of the CNN and GRU models were similar to those obtained at 11:00. However, the inversion effect at 15:00 [see Fig. 8(c)] was inferior to that at 11:00 and 13:00 due to decreased spectral reflectance caused by weakened light intensity, resulting in poor spectral reflectance of different OFTs. Based on the aforementioned analysis, the advantages of the CNN and GRU models in the inversion of different OFTs were demonstrated.

2) Quantitative Analysis of the Inversion Results: Quantitative evaluation of the effectiveness of each model in OFT inversion was conducted using the established evaluation metrics. The quantitative evaluation results for the three moments were presented in Table I. Given the large range of OFT in the experimental design, we partitioned the OFT into three sections for analysis. The inversion accuracy of the model was the mean of the five inversion results.

Based on the quantitative evaluation results, it could be observed that the CNN and GRU models had their respective advantages in inverting OFT in different thickness ranges compared to other models. Specifically, for OFT in the range of 0.01-0.30 mm, the 11:00 GRU model had an RMSE of 0.022 mm and an MRE of 0.195, which exceeded the range of the set OFT. The 13:00 GRU model had an RMSE of 0.015 mm and an MRE of 0.085, while the 15:00 GRU model had an RMSE of 0.034 mm and an MRE of 0.528. In this thickness range, the GRU model outperformed the SVR and CNN models in terms of RMSE and R^2 at 13:00 and 15:00.

For OFT in the range of 0.40-0.90 mm, the CNN model exhibited advantages over the other two models at 13:00 and 15:00. Specifically, the CNN model had an RMSE of 0.054 mm and an MRE of 0.047 for the 11:00 inversion accuracy, while the RMSE was 0.323 mm and the MRE was 0.083 for the thickness range of 1.00-3.04 mm, both within the set thickness range. The 13:00 inversion results showed that the model's accuracy increased with the increase in illumination intensity. The RMSE was 0.011 mm and the MRE was 0.009 for the thickness range of 0.40-0.90 mm, while the RMSE was 0.037 mm and the MRE was 0.018 for the thickness range of 1.00-3.04 mm. However, the inversion results for OFT became worse at 15:00 as the illumination intensity weakened. The RMSE was 0.088 mm and the MRE was 0.079 for the thickness range of 0.40-0.90 mm, while the RMSE was 0.310 mm and the MRE was 0.086 for the thickness range of 1.00-3.04 mm.

C. OFT Inversion Using 1DCNN_GRU Model

To address the issue of poor spectral separability of different OFTs, we proposed the 1DCNN_GRU model. Integrating CNN with the GRU model, the 1DCNN_GRU model utilized the 1DCNN model for feature extraction, followed by mapping the features into a sequence vector input for the GRU model. This approach allowed the model to remember different batches of spectral feature information, thereby achieving better inversion results during the training process. Through both visualization of the OFT grading chart and quantitative evaluation, we compared the inversion results of the proposed model with these of basic inversion model. Furthermore, this model was employed to explore the optimal timing for OFT inversion.

1) OFT Inversion Results: Taking into account the thickness classification maps from the three moments, it could

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TABLE II
ACCURACY EVALUATION OF OFT INVERSION RESULTS OF 1DCNN_GRU

OFT / Model	Accuracy Evaluation Indicators	11:00				13:00				15:00			
		1DCNN_GRU	lower than SVR	lower than CNN	lower than GRU	1DCNN_GRU	lower than SVR	lower than CNN	lower than GRU	1DCNN_GRU	lower than SVR	lower than CNN	lower than GRU
0.01-0.30mm	RMSE/mm	0.016	0.008	0.006	0.013	0.006	0.010	0.009	0.006	0.020	0.017	0.014	0.005
	MAE/mm	0.009	0.001	0.003	0.010	0.003	0.002	0.000	0.005	0.012	0.006	0.007	0.007
	MRE	0.195	0.000	0.000	0.077	0.084	0.029	0.001	0.138	0.218	0.244	0.310	0.100
	R^2	0.972	0.030	0.024	0.093	0.998	0.024	0.020	0.013	0.961	0.116	0.089	0.022
0.40-0.90mm	RMSE/mm	0.027	0.036	0.027	0.071	0.009	0.022	0.002	0.013	0.022	0.081	0.066	0.052
	MAE/mm	0.017	0.010	0.013	0.061	0.005	0.015	0.000	0.009	0.014	0.046	0.033	0.041
	MRE	0.027	0.014	0.020	0.082	0.009	0.024	0.000	0.014	0.024	0.073	0.055	0.054
	R^2	0.974	0.126	0.083	0.587	0.997	0.032	0.001	0.014	0.981	0.565	0.365	0.267
1.00-3.04mm	RMSE/mm	0.122	0.326	0.201	0.153	0.037	0.058	0.000	0.045	0.132	0.308	0.178	0.300
	MAE/mm	0.074	0.097	0.078	0.157	0.014	0.032	0.001	0.044	0.077	0.117	0.057	0.245
	MRE	0.044	0.026	0.039	0.076	0.018	0.008	0.000	0.011	0.042	0.059	0.044	0.105
	R^2	0.969	0.449	0.227	0.215	0.997	0.017	0.000	0.012	0.963	0.433	0.206	0.409

be concluded that for thinner oil films, the 1DCNN_GRU model solved the problem of unclear thickness classification and misinterpretation of thin oil films as thick ones in the CNN model, as well as the misclassification and overestimation of OFT in the SVR and GRU models. At 11:00, some improvements were made for the issue of the GRU model inverting OFT exceeding 3.5 mm. At 13:00, the 1DCNN_GRU model addressed the issue of misinterpreting thin oil films as thick ones in the CNN model at 1.50 mm thickness, and some improvements were made for the issue of misinterpreting thin oil films as thick ones in the CNN model at 0.04 and 0.30 mm thickness. Thus, the model could effectively resolve the problem of misclassification of thickness arising from poor spectral separability.

The inversion results from the three moments indicated that the 1DCNN_GRU model performed better than the other three models. At 11:00, the 1DCNN_GRU model showed improved performance in terms of RMSE, MAE, and R^2 compared to the other three models in the range of 0.01–0.03 mm OFT, but the improvement of MRE was not significant. For OFT exceeding 0.30 mm, the 1DCNN_GRU model outperformed the other three models, with an MRE improvement of 0.082 compared to the GRU model at 0.40–0.90 mm thickness, and an MRE improvement of 0.039 compared to the CNN model. At thicknesses ranging from 1.00 to 3.04 mm, the 1DCNN_GRU model exhibited an MRE improvement of 0.076 compared to the GRU model. At 13:00, although there was not a significant improvement in the MAE and MRE evaluation metrics compared to the CNN model, the 1DCNN_GRU model showed some improvements in RMSE and R^2. As for the analysis of the experimental results at 15:00, the 1DCNN_GRU model demonstrated even more significant improvements, especially when the spectral separability was reduced and the spectral features were not clear. This highlighted the advantage of integrating the two models. This model was not only capable of obtaining high-precision inversion results at times of good spectral separability, but also suitable for thickness inversion at times of poor spectral separability. From the results in Table II, it could be seen that the R^2 improved compared to the other three models. In the range of 0.01–0.30 mm thickness, the 1DCNN_GRU model outperformed the SVR model with an MRE improvement of 0.244, the CNN model with an MRE improvement of 0.310, and the GRU model with an MRE improvement of 0.100.

2) Optimal Time for OFT Inversion: Comparing the inversion results of 1DCNN_GRU at the three moments, we observed that the OFT inversion accuracy was highest at 13:00. In the OFT range of 0.01–0.30 mm, the RMSE was 0.01 mm higher than at 11:00, and the MRE increased by 0.111. At 0.40–0.90 mm OFT, the RMSE was improved by 0.018 mm compared to 11:00, and the same thickness was also improved at 1.00–3.04 mm. Based on the OFT grading chart at the three moments in Fig. 9, it was observed that the OFT classification was most apparent at 13:00. Compared to the inversion results at 11:00, there was no misclassification of thin oil films into thick ones for thicker oil films. Compared to the inversion results at 15:00, there was no issue of excessive thickness inversion. We believe that these phenomena were due to the influence of light intensity. With decreasing light intensity, the spectral reflectance of different OFTs decreased, and spectral separability deteriorated, resulting in decreased OFT inversion accuracy. Therefore, we recommend selecting the best OFT detection time based on hyperspectral data during noon with high light intensity.

IV. DISCUSSION

A. Detectable OFT Range of Airborne S185 Sensor

In the application of hyperspectral data for quantitative OFT inversion, its advantages of high resolution and a large number of bands can be utilized. However, it is unclear what is the optimal OFT range that can be detected by hyperspectral data. Therefore, in this section, we selected the OFT inversion results of the 1DCNN_GRU model at 13:00 for analysis. We plotted a scatter plot of the measured and inverted OFT values, and analyzed the quantitative evaluation of the individual OFT inversion results. In the scatter plot, we compared the inverted values with the measured values by plotting a 1:1 scale line and error lines.

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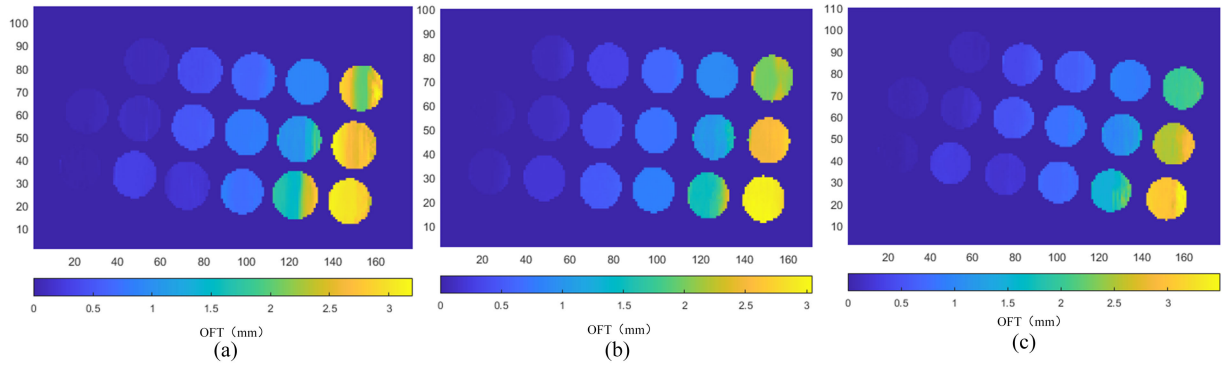


Fig. 9. OFT grading chart of 1DCNN_GRU at (a) 11:00, (b) 13:00, and (c) 15:00.

TABLE III
INVERSION ACCURACY OF EACH OFT USING HYPERSPECTRAL SENSOR

OFT/ mm	RMSE/ mm	MRE	OFT/ mm	RMSE/ mm	MRE
0.01	0.010	0.559	0.70	0.008	0.005
0.04	0.006	0.108	0.80	0.013	0.009
0.07	0.004	0.034	0.90	0.015	0.007
0.10	0.005	0.040	1.01	0.052	0.030
0.20	0.006	0.014	1.50	0.039	0.019
0.30	0.021	0.034	2.00	0.028	0.011
0.40	0.015	0.024	2.50	0.062	0.015
0.50	0.013	0.016	3.04	0.103	0.020
0.60	0.007	0.009			

The comparison between the inversion results and corresponding measurements of 17 OFTs ranging from 0.01 to 3.04 mm was displayed in Fig. 10(a). The figure also included error lines of 0.10 and 0.30 mm. Most of the inversion results were located on either side of the 1:1 scale line, with several results immediately adjacent to it. By combining Fig. 10(a) and (c), we observed that the OFT values ranging from 0.10 to 0.80 mm were all located within the acceptable error line of 0.10 mm. The results of MRE in Table III indicate that the average relative error did not exceed the set OFT, and the MRE of the oil film reached 0.005 at a thickness of 0.70 mm. The inversion accuracy of relatively thin oil films was found to be inferior to that of relatively thick ones within this thickness range.

The inversion results for OFT in the range of 0.01–0.30mm were displayed in Fig. 10(b), with error lines of 0.01 and 0.04 mm. The oil films falling within this thickness range exceeded the 0.01 mm error line. By combining the results in Table II, we observed that the MRE of oil films with a thickness of 0.01 mm reached 0.559, and the MRE of 0.04 mm was 0.108, both exceeding the set OFT. The poor accuracy of relatively thin inversion can be attributed to two factors. First, it was mainly due to significant errors in the experimental preparation of the oil film. Second, in conjunction with the analysis of spectral reflectance in Section III-A, it was discovered that the sensitivity of the airborne hyperspectral imaging system utilized in the experiment was inadequate to differentiate very thin oil films.

The OFT range of 0.90–3.04 mm exhibited some inversion results exceeding the 0.10 mm error lines, but the majority of it fell within the 0.30 mm error lines. However, the OFT value of 3.04 mm exceeded the 0.30 mm error lines. Fig. 10(d) plots the inversion results for the OFT range of 1.01–3.04 mm, with 0.50 mm error lines included. For an OFT of 3.04 mm, only a few inversion results were located outside the 0.50 mm error lines. Furthermore, in conjunction with Table III, we observed that the MRE of the 3.04 mm OFT reached 0.020, indicating that the inversion results for thicknesses greater than 0.90 mm were within a reasonable error range, which is deemed acceptable.

The analysis of inversion results for a single OFT showed that, in terms of RMSE, except for the OFT of 0.01 mm which had an RMSE of 0.01 mm, the inversion results for all other OFT were smaller than the corresponding thicknesses set in the experiment. Combined with MRE, it was found that the MRE was less than 0.40 mm for OFT above 0.07 mm, and the relative error was even smaller for relatively thick oil films. In summary, based on the inversion results for all OFT set in the experiment, it could be concluded that hyperspectral imaging had good detection performance for OFT greater than 0.07 mm, and particularly for OFTs of 0.60–0.90 mm. In future experiments, a wider range of OFT will be set to investigate the upper limit of the optimal OFT that can be detected by hyperspectral sensor.

B. Inversion and Evaluation of OFT Using HSI Assisted by BT Data

Previous research had indicated that thermal infrared technology performs well in detecting OFT greater than 0.40 mm [19]. In this study, we found that hyperspectral sensor had good performance in detecting OFT greater than 0.07 mm. Therefore, in this section, we conducted inversion and evaluation of OFT using HSI assisted by brightness temperature (BT) data. The two types of data were fused at both the feature level and decision level. Furthermore, we compared the differences between using single-source data and multisource data for inverting OFT, as well as the differences between feature level and decision level combinations for inverting OFT. The objective was to explore the advantages of combining these two technologies for detecting OFT.

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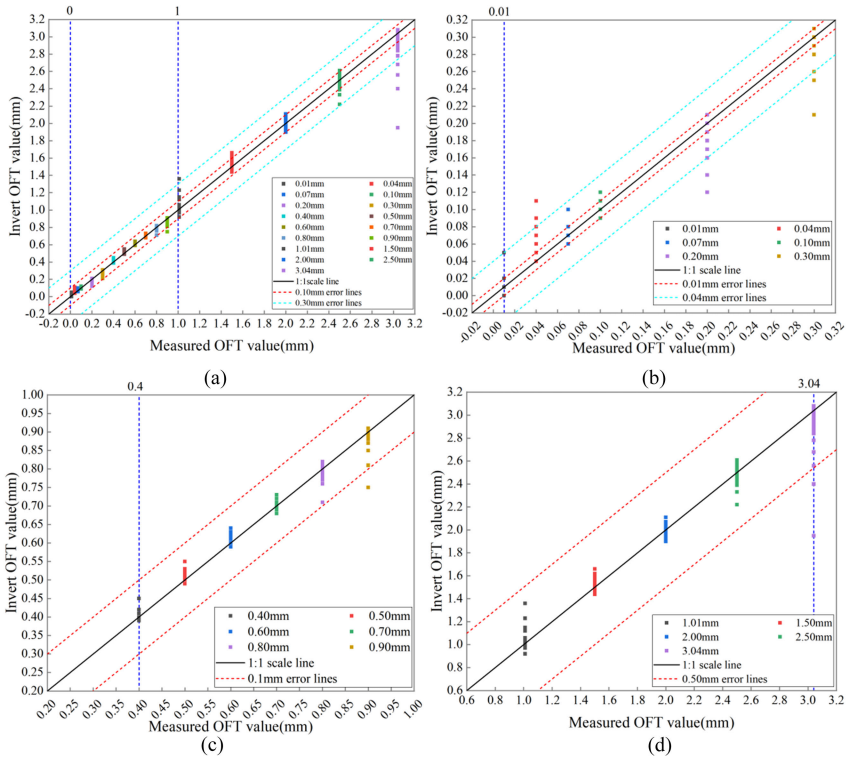


Fig. 10. Comparison of measured and inverted values of (a) 0.01–3.04 mm, (b) 0.01–0.30 mm, (c) 0.40–0.90 mm, and (d) 1.01–3.04 mm OFT.

1) Preprocessing of Hyperspectral and Thermal Infrared Data: In this section, the research focused on OFT monitoring using thermal infrared sensor, as different OFTs display different chiaroscuro on the thermal infrared image, which can be utilized to monitor the OFT. Previous research in [19] indicated that the best OFT detection time based on thermal infrared was at noon. Therefore, a thermal infrared BT image was obtained at the same time (13:00) as the HSIs, using a Testo 890-2 thermal infrared sensor with a spectral range of 8–14 μm, sensitivity NETD $\leq$ 0.04 K, and a photograph resolution of 640 $\times$ 480. The acquired thermal infrared BT image was corrected using meteorological data.

Before synthesizing the data, the acquired thermal infrared images needed to be registered, resampled, and cropped to match the resolution of the HSIs. After cropping both images to the same size, labels were created for the sample. To display the OFT grading chart of the inversion results, most of the area in the ring was chosen as the experimental area. Each ring consisted of approximately 190 pixels, and the specific experimental area was illustrated in Fig. 11. The selected experimental area was reasonable and effective as there was no shadow interference on the thermal infrared and HSIs within the chosen area.

2) Decision Fusion Algorithm Based on Fuzzy Membership: Decision-level fusion enhances decision-making accuracy, reliability, and efficiency. This is achieved by combining information from multiple sources. The limitations of a single sensor or information source are overcome. In this article, a decision-level fusion algorithm based on fuzzy membership was employed to fuse the inversion results obtained from the

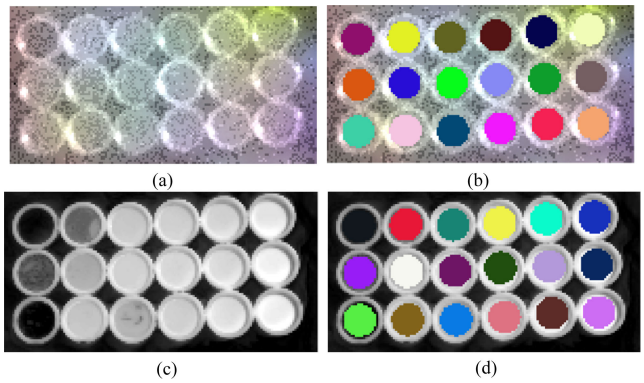


Fig. 11. (a) Cropped hyperspectral true color image. (b) Corresponding label map for the cropped hyperspectral true color image. (c) Resampled thermal infrared image. (d) Corresponding label map for the resampled thermal infrared image.

two data sources, namely hyperspectral and thermal infrared data. Prior to decision fusion, the OFT needed to be segmented

$$T_r = \begin{cases} (-\infty, T_n + 0.005), n = 1 \\ (T_n - 0.005, T_n + 0.015), n = 2 \\ (T_n - 0.015, T_n + 0.015), 3 \leq n \leq 4 \\ (T_n - 0.015, T_n + 0.05), n = 5 \\ (T_n - 0.05, T_n + 0.05), 6 \leq n \leq 13 \\ (T_n - 0.05, T_n + 0.25), n = 14 \\ (T_n - 0.25, T_n + 0.25), 15 \leq n \leq 17 \\ (T_n + 0.25, +\infty), n = 18 \end{cases} \tag{14}$$

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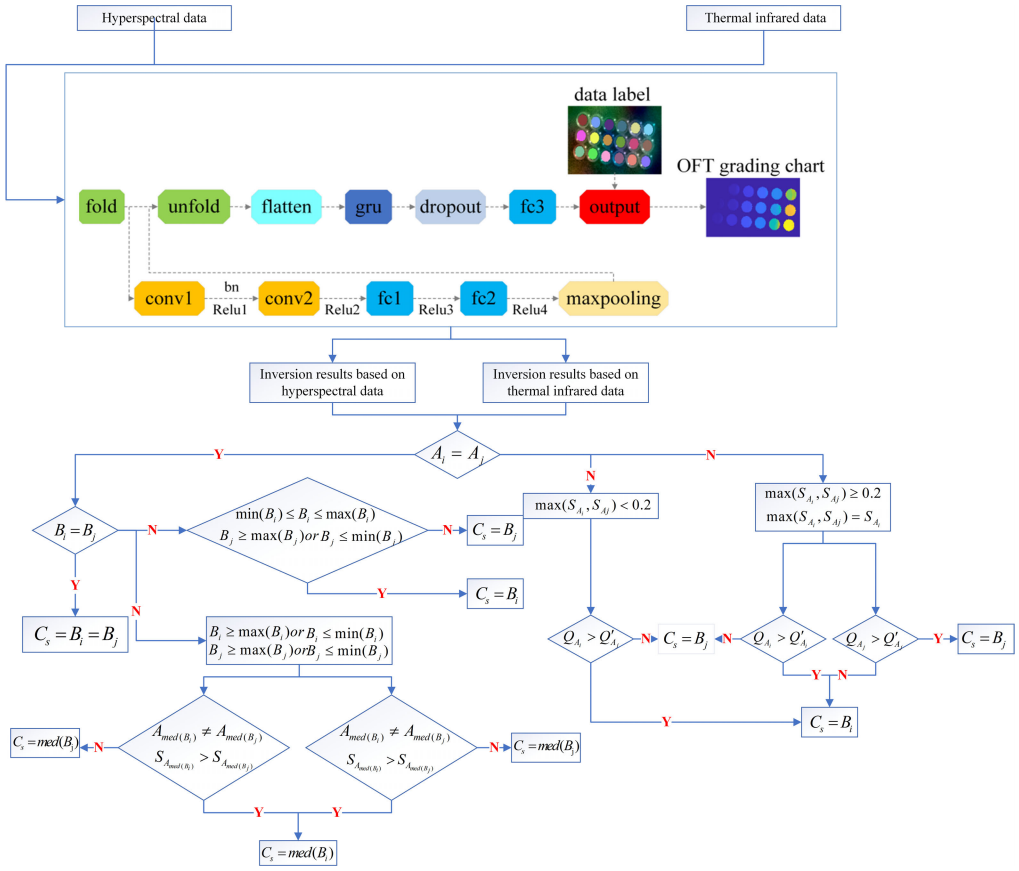


Fig. 12. Decision fusion process for OFT.

$$T_n = \begin{cases} 0.00, n = 1 \\ 0.01 + (n-2) \times 0.03, 2 \le n \le 5 \\ 0.1 + (n-5) \times 0.1, 6 \le n \le 14 \\ 1.0 + (n-14) \times 0.5, 15 \le n \le 18 \\ n = 1, 2, 3, \ldots 18 \end{cases} \tag{15}$$

where T_r represents the range of OFT for different OFT segments, and T_n represents the value of OFT.

To perform decision-level fusion, we first calculated the average relative accuracy of each OFT inversion result and the fuzzy membership of the OFT segment. The formula for calculating the membership of pixel m belonging to the OFT segment k is shown in the following equation, denoted as S_k^m [47]

$$S_k = 0.5 + \left\{ \sum_{i=1}^{n} w_j (S_k^m - 0.5)^{\alpha} \right\}^{\frac{1}{\alpha}} \tag{16}$$

where S_k represents the degree of membership of a single OFT segment. The range of k is [1], [18]. w_j represents the relative importance of each feature, and $\sum_{j=1}^{n} w_j = 1$. n represents the number of results to be fused. α is an odd number. And $w_1 = (a/a+b)$ $w_2 = (b/a+b)$. The values of a and b are determined based on the average relative accuracy of the inversion results obtained from hyperspectral spectral reflectance data and thermal infrared brightness temperature difference (BTD) data, respectively.

Decision fusion mainly considers two situations: (1) when the OFT segment of the inversion result is the same for both hyperspectral and thermal infrared data, (2) and when the OFT segment of the inversion result is different for hyperspectral and thermal infrared data. i and j represent hyperspectral and thermal infrared data, respectively. A_i and A_j represent the OFT segments where the hyperspectral and thermal infrared inversion results are located, respectively. B_i and B_j represent hyperspectral and thermal infrared inversion OFT values, respectively. $\max(\cdot)$, $\min(\cdot)$, $\mathrm{med}(\cdot)$ represent the maximum, minimum, and median values of the 5×5 fields centered on the current pixel. Q_A and Q'_{A_j} represent the average relative accuracy of the OFT segment in which the cell is located, respectively. S_{ai} and S_{aj} represent the fuzzy membership of the OFT segment in which the cell is located, respectively. C is the result of decision fusion. The decision fusion process for OFT inversion results is illustrated in Fig. 12.

3) Analysis of Joint Inversion Results: In this article, the 1DCNN_GRU model was proposed to invert the OFT. The input data for the model consisted of spectral reflectance data for HSIs and the BTD data between oil and water for thermal infrared images. To evaluate the ability of hyperspectral and thermal infrared sensor to invert the OFT, the thermal infrared data that had not been resampled was used. The feature level fusion of hyperspectral and thermal infrared data selected the normalized spectral reflectance data and the normalized BTD data. The fusion of normalized thermal infrared data and normalized hyperspectral data through spectral band

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TABLE IV
INVERSION ACCURACY OF OFT BASED ON SINGLE-SOURCE AND MULTISOURCE DATA UNION

OFT / Data	Accuracy Evaluation Indicators	Hyperspectra	Thermal infrared	Hyperspectral and thermal infrared (feature level)	Hyperspectral and thermal infrared (decision level)
0.01-0.30mm	RMSE/mm	0.010	0.005	0.006	0.013
	MAE/mm	0.005	0.003	0.003	0.007
	MRE	0.183	0.042	0.098	0.237
	R^2	0.989	0.997	0.997	0.982
0.40-0.90mm	RMSE/mm	0.012	0.021	0.013	0.012
	MAE/mm	0.007	0.015	0.008	0.007
	MRE	0.012	0.025	0.013	0.011
	R^2	0.995	0.983	0.994	0.995
1.00-3.04mm	RMSE/mm	0.065	0.087	0.064	0.045
	MAE/mm	0.037	0.058	0.035	0.025
	MRE	0.019	0.033	0.018	0.013
	R^2	0.991	0.985	0.992	0.996

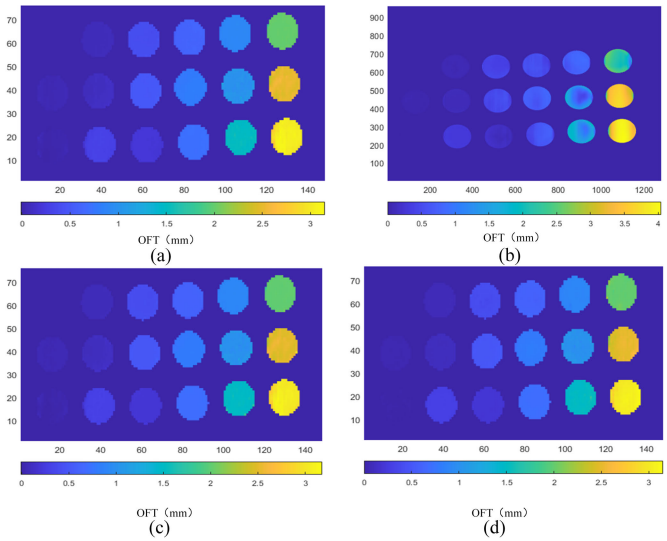


Fig. 13. OFT grading diagram of (a) hyperspectra, (b) thermal infrared, (c) fusion of hyperspectral and thermal infrared decision level, and (d) fusion of hyperspectral and thermal infrared characteristic level.

TABLE V
INVERSION ACCURACY OF SINGLE OFT USING HYPERSPECTRAL AND THERMAL INFRARED

Data	Hyperspectra		Thermal infrared	
OFT/mm	RMSE/mm	MRE	RMSE/mm	MRE
0.01	0.010	0.559	0.003	0.080
0.04	0.006	0.108	0.040	0.050
0.07	0.004	0.034	0.008	0.077
0.10	0.005	0.040	0.010	0.063
0.20	0.006	0.014	0.008	0.028
0.30	0.021	0.034	0.007	0.011
0.40	0.015	0.024	0.016	0.030
0.50	0.013	0.016	0.017	0.028
0.60	0.007	0.009	0.021	0.029
0.70	0.008	0.005	0.018	0.029
0.80	0.013	0.009	0.017	0.010
0.90	0.015	0.007	0.027	0.017
1.01	0.052	0.030	0.092	0.050
1.50	0.039	0.019	0.079	0.040
2.00	0.028	0.011	0.073	0.029
2.50	0.062	0.015	0.094	0.029
3.04	0.103	0.020	0.176	0.031

concatenation was performed, serving as input data for feature-level fusion. The output data was the measured OFT value. The input data for decision fusion consisted of the oil film inversion results obtained from hyperspectral and thermal infrared data. Randomly selected 60% of the samples were used for training, and all samples were used for testing. The inversion results were averaged over five runs and evaluated using thickness segmentation (see Table IV) and individual thicknesses (see Table V). A thickness grading plot of the OFT inversion results was generated (see Fig. 13).

The results obtained from the inversion of original thermal infrared data were compared to those obtained from hyperspectral inversion, and the results were presented in Table IV and the thickness inversion charts. The findings revealed that for OFT ranging from 0.01 to 0.30 mm, the inversion results of thermal infrared surpassed those of hyperspectral data, while for OFT in the range of 0.40–3.04 mm, hyperspectral demonstrated a more apparent advantage. In comparison, hyperspectral was more accurate in inverting the OFT. Furthermore, upon comparing the inversion results of thermal infrared for each thickness as shown in Table IV, it was discovered that the inversion accuracy for OFT in the range of 0.20–3.04 mm was superior to that of OFT in the range of 0.01–0.10 mm. These findings were consistent with the conclusions reached in [17], which indicated that thermal infrared was more advan-

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tageous in detecting thicker oil films. According to the results obtained in Section IV-A, hyperspectral was more beneficial in detecting OFT greater than 0.07 mm. Therefore, it was necessary to combine hyperspectral and thermal infrared data to detect the OFT. The primary approach involved utilizing the advantage of thermal infrared compared to hyperspectral data in the thickness range of 0.01–0.30 mm to enhance the inversion accuracy of the spectrum. Following this, the detection advantage of thermal infrared itself was employed to improve the joint inversion accuracy for thicker oil films.

The results of feature-level fusion indicate that there was a certain improvement in OFT inversion accuracy compared to the results obtained from single-source data. However, as indicated in Table IV, the enhancement in accuracy was not significant. For instance, within the thickness range of 0.10–3.04 mm, the MRE was only 0.001 mm higher than hyperspectral, the RMSE increased by 0.001 mm, and there was hardly any substantial change in the 0.40–0.90 mm thickness range. Nevertheless, it was observed that the results of the higher MRE spectrum improved by 0.085 in the thickness range of 0.01–0.30 mm. The feature-level fusion results indicated that the addition of thermal infrared bands improved the inversion accuracy of hyperspectral in the thickness range of 0.01–0.30 mm.

The decision-level fusion results showed an improvement in the inversion results compared to single-source data in the thickness range of 0.40–3.04 mm. Notably, in the thickness range of 1.01–3.04 mm, the RMSE increased by 0.020 mm and the MRE increased by 0.006. The principle of this algorithm was to preserve excellence by using certain rules to retain the superior results. Therefore, only the optimal results were retained. Upon comparing the inversion results of the feature-level and decision-level fusion, it was discovered that the feature level had a significant improvement effect in the first OFT segment, while decision-level fusion demonstrated significant improvement in the third OFT segment, compared to single-source data. The experiments demonstrated that the inversion accuracy was enhanced after the union of the two data sources.

In real-world oil spill scenarios, the detection of the OFT can be achieved by combining hyperspectral and thermal infrared data, resulting in improved accuracy of OFT detection. However, the actual oil spill scenario environment is complex, and there may be certain challenges in applying the experimental conclusions obtained under ideal experimental conditions to real-world oil spill scenarios. In the forthcoming experiments, we plan to include more complex environmental variables to investigate the impact of joint detection under different environmental conditions.

V. Conclusion

This study utilized airborne hyperspectral sensor to obtain images of different OFTs through a ground experiment, analyzed the spectral characteristics of different OFTs, and developed a model to quantitatively invert OFT. The optimal range of OFT that can be detected by airborne hyperspectral sensor was also explored. Additionally, to investigate the ability of multisource data fusion for inversion of OFT, thermal infrared sensor was introduced, and the two types of data were combined at both the feature level and decision level. The differences between multisource data fusion and single-source data inversion of OFT were compared.

Based on the questions raised earlier, this article reaches the following main conclusions. 1) Based on the analysis of the spectral curves, the airborne S185 hyperspectral sensor can be used for OFT monitoring, but for oil films thicker than 0.70 mm, there might be poor spectral separability for similar OFT. 2) According to the advantages of the CNN and GRU models, the 1DCNN_GRU model was proposed in this article for inverting OFT. It was found that the model could accurately invert different OFTs, and the inversion results outperformed three basic models, including SVR, CNN, and GRU. Especially at 15:00 when spectral separability was poor, for the three thickness ranges of 0.01–0.30, 0.40–0.90, and 1.00–3.04 mm, compared to the three basic models, the MRE of the proposed model was, respectively, reduced by 0.100–0.310, 0.052–0.081, and 0.044–0.105. Moreover, the optimal time for inverting OFT using hyperspectral sensors was around noon. 3) Through the analysis of the experimental results at 13:00, this article found that airborne hyperspectral sensors performed well in detecting OFT greater than 0.07 mm, especially for those within the 0.60–0.90 mm range. 4) The joint use of thermal infrared and hyperspectral data was superior to single-source data in inverting OFT. Especially, fusing the data at the feature level can improve the inversion accuracy for OFTs of 0.01–0.30 mm. For OFTs of 1.00–3.04 mm, decision-level fusion led to better inversion accuracy in that range.

The current experiment was conducted during autumn in a controlled laboratory environment. However, the actual marine environment is complex and constantly changing. Therefore, further experiments are necessary to verify the effectiveness of the model, and explore the OFT detection capability of hyperspectral remote sensing technology under different environmental variables. It can provide a more comprehensive and accurate scientific basis for the application of hyperspectral remote sensing technology in marine oil spill emergency response. Furthermore, to further improve the accuracy of OFT inversion, selecting appropriate data fusion methods for different thickness ranges is the next research direction.

Acknowledgment

The authors thank the Editor-in-Chief, the Associate Editor, and the reviewers for their insightful comments and suggestions.

References

[1] J. Beyer, H. C. Trannum, T. Bakke, P. V. Hodson, and T. K. Collier, "Environmental effects of the Deepwater Horizon oil spill: A review," *Mar. Pollut. Bull.*, vol. 110, no. 1, pp. 28–51, 2016.

[2] Q. Zhu et al., "Oil spill contextual and boundary-supervised detection network based on marine SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5213910.

[3] A. Dhaka and P. Chattopadhyay, "A review on physical remediation techniques for treatment of marine oil spills," *J. Environ. Manag.*, vol. 288, no. 15, p. 112428, 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:17:39 UTC from IEEE Xplore. Restrictions apply.

[4] D. R. Mishra et al., "Post-spill state of the marsh: Remote estimation of the ecological impact of the Gulf of Mexico oil spill on Louisiana Salt Marshes," *Remote Sens. Environ.*, vol. 118, no. 15, pp. 176–185, 2012.

[5] J. Wang, X. Liu, Y. Zhang, and G. Han, "A review of oil spill remote sensing," *Remote Sens.*, vol. 9, no. 11, p. 1091, 2017.

[6] G. Liu, J. Li, Y. Zhang, and H. Wu, "An oil spill detection algorithm for high-resolution optical remote sensing images based on adaptive threshold segmentation," *Remote Sens.*, vol. 11, no. 7, p. 776, 2019.

[7] X. Li, J. Chen, and X. Li, "Oil spill detection using synthetic aperture radar and optical remote sensing: A review," *Remote Sens.*, vol. 11, no. 5, p. 556, 2019.

[8] J. D. Runnels and S. A. Socolofsky, "Deepwater Horizon oil spill: A review of the planktonic response," *J. Mar. Res.*, vol. 73, no. 1, pp. 7–39, 2015.

[9] S. Han, J. Lee, and M. Lee, "An acoustic monitoring system for oil spill detection and classification," *Sensors*, vol. 17, no. 7, p. 1619, 2017.

[10] S. Liu, X. Zhao, Z. Li, Y. Wang, and X. Zhang, "A review of oil spill remote sensing technology," *J. Oceanol. Limnol.*, vol. 37, no. 5, pp. 1435–1446, 2019.

[11] Y. Li and Z. Li, "An unmanned aerial vehicle based approach for oil spill detection and monitoring," *Remote Sens.*, vol. 10, no. 2, p. 306, 2018.

[12] Y. Zhang, L. Wang, and X. Li, "A new oil spill detection method based on GF-3 quad-polarization SAR imagery," *Remote Sens.*, vol. 10, no. 11, p. 1726, 2018.

[13] Q. Zhu et al., "Oil spill contextual and boundary-supervised detection network based on marine SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–10, 2022, Art. no. 5213910.

[14] L. Guo, J. Wang, and Y. Cheng, "Research on the detection of oil slick thickness by synthetic aperture radar," *J. Ocean. U China*, vol. 17, no. 2, pp. 269–276, 2018.

[15] Y. Lu, W. Zhan, and C. Hu, "Detecting and quantifying oil slick thickness by thermal remote sensing: A ground-based experiment," *Remote Sens. Environ.*, vol. 181, pp. 207–217, Aug. 2016.

[16] J. Jiao, Y. Lu, C. Hu, J. Shi, S. Sun, and Y. Liu, "Quantifying ocean surface oil thickness using thermal remote sensing," *Remote Sens. Environ.*, vol. 261, Aug. 2021, Art. no. 112513.

[17] Z. Jiang, Y. Ma, J. Zhang, X. Mao, K. Du, and L. Zhang, "Research on thermal infrared remote sensing detection of oil spill on sea surface," in *Proc. IGARSS*, Kuala Lumpur, Malaysia, 2022, pp. 7970–7973.

[18] M. Wang, J. Yang, S. Liu, J. Wan, and J. Zhang, "Thermal infrared detection of oil film thickness at sea: Airborne and portable thermal imagers are used in one experiment," in *Proc. IGARSS*, Pasadena, CA, USA, 2023, pp. 1853–1856.

[19] M. Wang, J. Yang, S. Liu, J. Zhang, Y. Ma, and J. Wan, "Quantitative inversion ability analysis of oil film thickness using bright temperature difference based on thermal infrared remote sensing: A ground-based simulation experiment of marine oil spill," *Remote Sens.*, vol. 15, no. 8, p. 2018, Apr. 2023.

[20] X. Li, J. Zhu, and Y. Zhou, "Oil spill detection in the Bohai Sea using thermal infrared imagery: Comparison of two algorithms," *Int. J. Remote Sens.*, vol. 39, no. 6, pp. 1827–1847, 2018.

[21] L. E. Sweet et al., "Exposure to ultraviolet radiation late in development increases the toxicity of oil to mahi-mahi (*Coryphaena hippurus*) embryos," *Environ. Toxicol. Chem.*, vol. 36, pp. 1592–1598, Jun. 2017.

[22] Z. Suo et al., "Ultraviolet remote sensing of marine oil spills: A new approach of Haiyang-1C satellite," *Opt. Exp.*, vol. 29, pp. 13486–13495, 2021.

[23] Q. Zhu, Y. Zhang, and Z. Hu, "Detection of oil slick thickness using airborne laser-induced fluorescence spectroscopy," *Appl. Opt.*, vol. 57, no. 28, pp. 8278–8283, 2018.

[24] J. Wang, X. Wang, and X. Li, "Oil spill detection using LiDAR remote sensing technology: A case study in the Bohai Sea, China," *Remote Sens.*, vol. 11, no. 9, p. 1088, 2019.

[25] Y. Li, Q. Wu, and Y. Chen, "Oil spill detection using terrestrial laser scanning: A review," *Remote Sens.*, vol. 11, no. 9, p. 1031, 2019.

[26] T. Hu, J. Yuan, X. Wang, C. Yan, and X. Ju, "Spectral–spatial features extraction of hyperspectral remote sensing oil spill imagery based on convolutional neural networks," *IEEE Access*, vol. 10, pp. 127969–127983, 2022.

[27] J. Yang, Y. Hu, Y. Ma, Z. Li, and J. Zhang, "Research on oil spill pollution type identification using RPnet deep learning model and airborne hyperspectral image," in *Proc. IGARSS*, Kuala Lumpur, Malaysia, 2022, pp. 807–810.

[28] L. Guo, Z. Wang, C. Wu, and Y. Ding, "Hyperspectral remote sensing for oil spill detection: A review," *J. Ocean. U China*, vol. 18, no. 4, pp. 763–772, 2019.

[29] X. Kang, B. Deng, P. Duan, X. Wei, and S. Li, "Self-supervised spectral–spatial transformer network for hyperspectral oil spill mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507410.

[30] H. Zhang, Z. Li, X. Zhang, and R. Liu, "A high-resolution hyperspectral method for oil thickness detection," *Remote Sens.*, vol. 9, no. 11, p. 1145, 2017.

[31] Z. Li, X. Zhang, H. Zhang, and R. Liu, "Oil spill thickness estimation using hyperspectral data and a modified Gaussian model," *Remote Sens.*, vol. 10, no. 8, p. 1236, 2018.

[32] J. Zhang, H. Li, and W. Zhang, "Oil thickness estimation using a modified Gaussian model and hyperspectral data," *Remote Sens.*, vol. 11, no. 16, p. 1905, 2019.

[33] X. Yang and J. Zhang, "A review on oil thickness estimation using remote sensing methods," *J. Ocean. U China*, vol. 18, no. 5, pp. 1049–1058, 2019.

[34] Y. Chen, J. Zhang, H. Li, and W. Zhang, "Oil slick thickness estimation based on Gaussian model and hyperspectral data," *J. Appl. Remote. Sens.*, vol. 10, no. 2, 2016, Art. no. 026018.

[35] Z. Jiang, J. Zhang, Y. Ma, and X. Mao, "Hyperspectral remote sensing detection of marine oil spills using an adaptive long-term moment estimation optimizer," *Remote Sens.*, vol. 14, no. 1, p. 157, Dec. 2021.

[36] Y. Wang, J. Zhang, and H. Li, "Oil spill thickness estimation using machine learning algorithms and hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3059–3071, 2020.

[37] Y. Wang, J. Zhang, H. Li, and W. Zhang, "Oil slick thickness estimation using convolutional neural networks and hyperspectral data," *Remote Sens.*, vol. 13, no. 7, p. 1317, 2021.

[38] H. Li, J. Zhang, and W. Zhang, "Oil slick thickness estimation based on machine learning and hyperspectral data," *Remote Sens.*, vol. 11, no. 7, p. 814, 2019.

[39] H. Li, J. Zhang, W. Zhang, and Y. Chen, "Oil slick thickness estimation using a random forest regression algorithm and hyperspectral data," *Remote Sens.*, vol. 10, no. 5, p. 723, 2018.

[40] Y. Hu, J. Zhang, Y. Ma, J. An, G. Ren, and X. Li, "Hyperspectral coastal wetland classification based on a multiobject convolutional neural network model and decision fusion," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 7, pp. 1110–1114, Jul. 2019.

[41] C.-C. Chang and C.-J. Lin, "LIBSVM: A library for support vector machines," *ACM Trans. Intell. Syst. Technol.*, vol. 2, no. 3, pp. 1–27, Apr. 2011.

[42] D. Huang et al., "Inversion prediction of COD in wastewater based on hyperspectral technology," *J. Cleaner Prod.*, vol. 385, Jan. 2023, Art. no. 135681.

[43] F. Liu, Z. Zhang, and R. Zhou, "Automatic modulation recognition based on CNN and GRU," *Tsinghua Sci. Technol.*, vol. 27, no. 2, pp. 422–431, Apr. 2022.

[44] S. Wang, K. Yang, Y. Shi, F. Yang, H. Zhang, and Y. Ma, "Prediction of over-the-horizon electromagnetic wave propagation in evaporation ducts based on the gated recurrent unit network model," *IEEE Trans. Antennas Propag.*, vol. 71, no. 4, pp. 3485–3496, Apr. 2023.

[45] B. X. Liu, "Extraction and analysis of oil film on water based on hyperspectral characteristics," Ph.D. dissertation, School Navigat. Dept., Dalian Maritime Univ., Dalian, China, 2013.

[46] P. Sun, M. P. Song, and J. B. An, "Study of prediction models for oil thickness based on spectral curve," *Spectrosc. Spectr. Anal.*, vol. 33, pp. 1881–1885, Jul. 2013.

[47] J. Yang et al., "Decision fusion of deep learning and shallow learning for marine oil spill detection," *Remote Sens.*, vol. 14, no. 3, p. 666, Jan. 2022.



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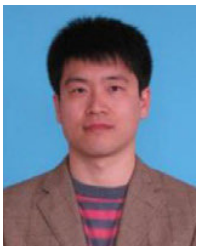
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